ECE341 HOMEWORK 3

Problem 1

Truth Table

D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0	1	2
0	0	1	1	1	1	1	1	0	0	1	3
0	1	0	0	0	1	1	0	0	1	1	4
0	1	0	1	1	0	1	1	0	1	1	5
0	1	1	0	1	0	1	1	1	1	1	6
0	1	1	1	1	1	1	0	0	0	0	7
1	0	0	0	1	1	1	1	1	1	1	8
1	0	0	1	1	1	1	1	0	1	1	9
1	0	1	0	1	1	1	0	1	1	1	A
1	0	1	1	0	0	1	1	1	1	1	b
1	1	0	0	1	0	0	1	1	1	0	C
1	1	0	1	0	1	1	1	1	0	1	d
1	1	1	0	1	0	0	1	1	1	1	E
1	1	1	1	1	0	0	0	1	1	1	F

Karnaugh Maps and Derived Expressions

For each K-map, the bits marked with diagonal lines correspond to the **first expression** shown below the K-map, while the bits **without** diagonal lines correspond to the **second expression** shown below it.

BA \ DC	00	01	11	10
00	X 1	X 1	X 0	X 1
01	X 1	X 0	X 1	X 1
11	X 1	X 1	X 0	X 0
10	X 1	X 0	X 0	X 1

$a = B\bar{D} + BC + A\bar{D}C + \bar{A}\bar{C} + \bar{A}\bar{B}D + \bar{B}D\bar{C}$

$b = \bar{C}\bar{D} + \bar{A}\bar{C} + \bar{A}\bar{B}\bar{D} + AB\bar{D} + A\bar{B}D$

BA \ DC	00	01	11	10
00	X 1	X 0	X 1	X 1
01	X 0	X 1	X 1	X 1
11	X 1	X 0	X 0	X 1
10	X 1	X 1	X 1	X 0

$c = \bar{B}\bar{D} + A\bar{D} + C\bar{D} + A\bar{B} + \bar{C}D$

$d = \bar{B}D + A\bar{B}C + \cancel{AD\bar{C}} + B\bar{A}C + \cancel{B\bar{D}\bar{C}} + \cancel{\bar{A}\bar{B}\bar{C}}$
$+ AB\bar{C}$ $\quad \bar{A}\bar{C}\bar{D}$

BA \ DC	00	01	11	10
00	X 1	X 1	X 1	X 1
01	X 0	X 1	X 0	X 1
11	X 0	X 0	X 1	X 1
10	X 0	X 1	X 1	X 1

$e = B\bar{A} + BD + CD + \cancel{\bar{A}\bar{B}\bar{C}}\ \bar{A}\bar{C}$

$f = \bar{A}\bar{B} + D\bar{C} + BD + \bar{B}\bar{D}C + \bar{A}\bar{C}$

BA \ DC	00	01	11	10
00	0	1	0	1
01	0	1	1	1
11	1	0	1	1
10	1	1	1	1

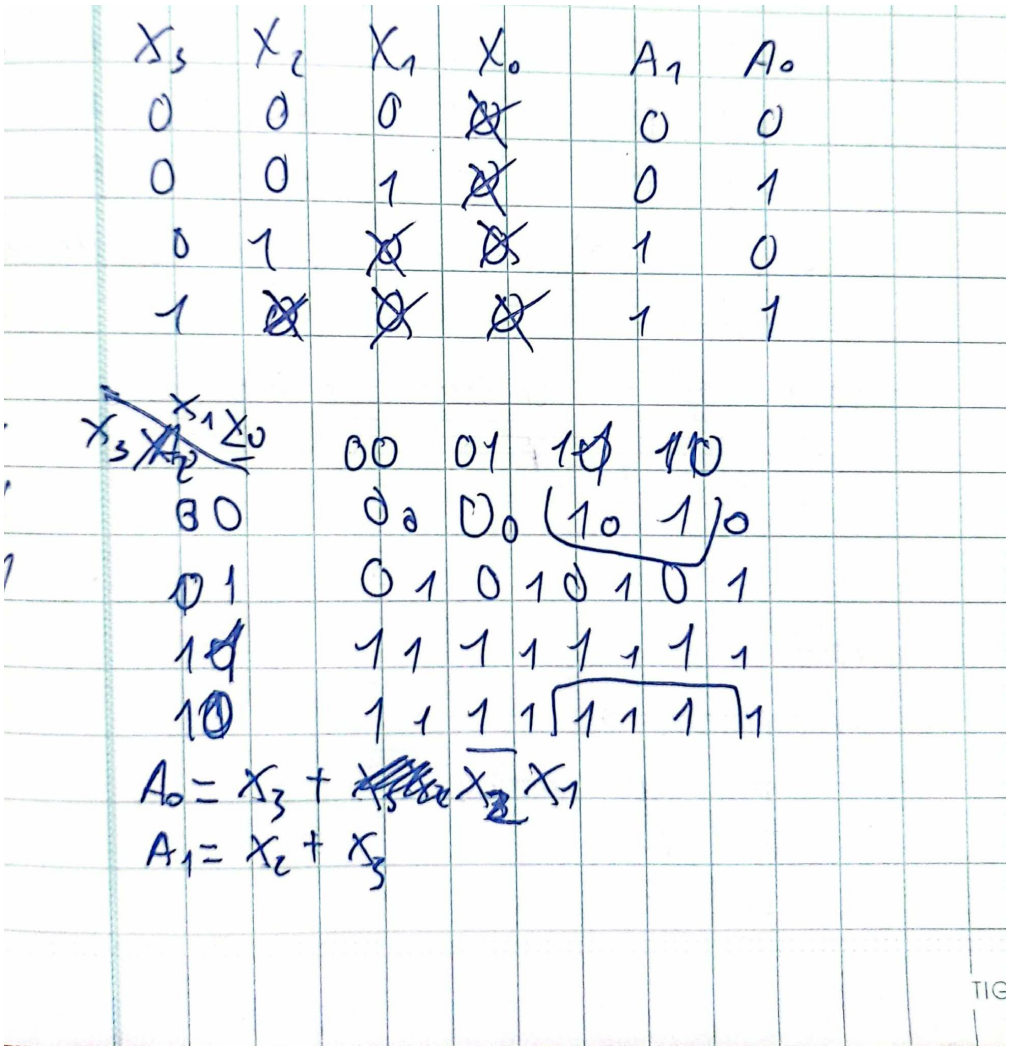
$g = B\bar{A} + B\bar{C} + AD + D\bar{C} + \bar{B}\bar{D}C$

Problem 2

Add `NOT` after each `OR` (`NOR` literally) in problem 1 and we are done.

Problem 3

Truth Table, K-map and Derived Expressions



X_3	X_2	X_1	X_0	A_1	A_0
0	0	0	X	0	0
0	0	1	X	0	1
0	1	X	X	1	0
1	X	X	X	1	1

$X_3 X_2$ \ $X_1 X_0$	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	1	1	1
10	1	1	1	1

$A_0 = X_3 + \overline{X_2} X_1$

$A_1 = X_2 + X_3$

Problem 4

We use one decoder to decide "which group of four" is active, and the other decoders to select the specific output within that group. Since a 4-bit input (A_3, A_2, A_1, A_0) has 16 possible states, we need five 2-to-4 decoders in total.

To decode 4 bits (A_3, A_2, A_1, A_0 where A_3 is the Most Significant Bit):

- The "Master" Decoder: Uses the two highest bits (A_3, A_2) to enable one of the four "Slave" decoders.
- The "Slave" Decoders: All four use the two lowest bits (A_1, A_0) to drive the final 16 outputs ($0-15$).

We now have 16 output pins across the four slave decoders:

- Slave 1 outputs represent Binary 0000 to 0011 (Decimal 0–3).
- Slave 2 outputs represent Binary 0100 to 0111 (Decimal 4–7).
- Slave 3 outputs represent Binary 1000 to 1011 (Decimal 8–11).
- Slave 4 outputs represent Binary 1100 to 1111 (Decimal 12–15).

Problem 5

First Stage (Select Input S_0): Use two 2:1 Muxes.

- Mux 1 handles data inputs X_0 and X_1.
- Mux 2 handles data inputs X_2 and X_3.
- Both share the same first select bit (S_0).

Second Stage (Select Input S_1): Use one final 2:1 Mux.

- This Mux takes the outputs from the first two Muxes as its data inputs.
- It uses the second select bit (S_1) to decide which "pair" of inputs to pass to the final output Q.

That design works because:

- When S_1 is 0, the 3rd Mux only cares about the "top half" (X_0 and X_1). S_0 then chooses between those two.
- When S_1 is 1, the 3rd Mux switches to the "bottom half" (X_2 and X_3). S_0 then chooses between those two.

Problem 6

4:1 Multiplexer (Mux)

A 4:1 Mux uses 2 select bits (S_1, S_0) to choose one of four data inputs (X_0, X_1, X_2, X_3).

Boolean Expression

$$Q = (X_0 \cdot \bar{S_1} \cdot \bar{S_0}) + (X_1 \cdot \bar{S_1} \cdot S_0) + (X_2 \cdot S_1 \cdot \bar{S_0}) + (X_3 \cdot S_1 \cdot S_0)$$

1:4 Demultiplexer (Demux)

A Demux performs the reverse: it takes one data input (X) and routes it to one of four outputs (Y_0, Y_1, Y_2, Y_3) based on the select lines (S_1, S_0).

Boolean Expressions

- $Y_0 = X \cdot \bar{S_1} \cdot \bar{S_0}$
- $Y_1 = X \cdot \bar{S_1} \cdot S_0$
- $Y_2 = X \cdot S_1 \cdot \bar{S_0}$
- $Y_3 = X \cdot S_1 \cdot S_0$

Enhancement for Expansion

To easily expand these into 8:1 or 1:8 versions, we add an **Enable (EN)** pin to the blocks. This follows the same hierarchical logic used in decoders.

For the Mux (8:1 Expansion)

- **The Logic:** Add the EN signal as a 4th input to every AND gate in your 4:1 Mux.
- **The Expansion:** To make an 8:1 Mux:
    1. Use two 4:1 Mux blocks.
    2. Connect a 1:2 Decoder (or a simple NOT gate) to the **Enable** pins using a 3rd select bit (S_2).

3. Feed the outputs of both 4:1 Muxes into a final **2-input OR gate**.

For the Demux (1:8 Expansion)

- **The Logic:** Similarly, add an EN signal as a 4th input to the AND gates.
- **The Expansion:** To make a 1:8 Demux:
  1. Use the 3rd select bit (S_2) to enable either "Demux Block A" (outputs 0-3) or "Demux Block B" (outputs 4-7).
  2. The data input D is sent to both blocks, but only the enabled block will pass the signal through.

Problem 7

We use output of the first multiplexer in the chip 74153 for part (a), and the second multiplexer for part (b).

Part (a)

Truth Table

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Transform the truth table to use B and C as inputs

B	C	Y
0	0	$\bar{A}$
0	1	0
1	0	1
1	1	1

Part (b)

Truth Table

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Transform the truth table to use B and C as inputs

B	C	Y
0	0	1
0	1	1
1	0	0
1	1	1